화학반응속도론 (2014년 2학기)
제3장 복잡반응계의 반응메커니즘

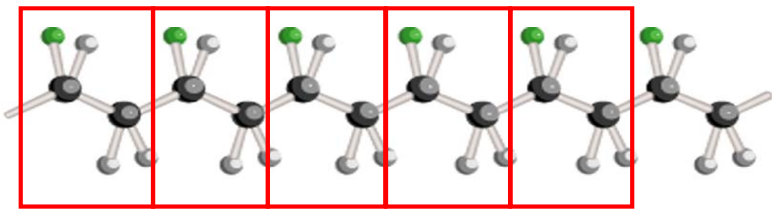
중요 학습 내용

- 중합반응: 단계중합반응과 연쇄중합반응
- 촉매작용반응과 Michaelis-Menten 메커니즘
- Lineweaver-Burk 도표

3.1 고분자와 중합반응

- 고분자 (polymer): 구조단위체인 단량체 (monomer)들의 공유결합에 의해 형성된 분자

Greek: "polys" ~ many
"meros"~parts



- 중합반응 (polymerization): 단량체 (monomer)들이 공유결합하여 고분자를 형성하는 반응



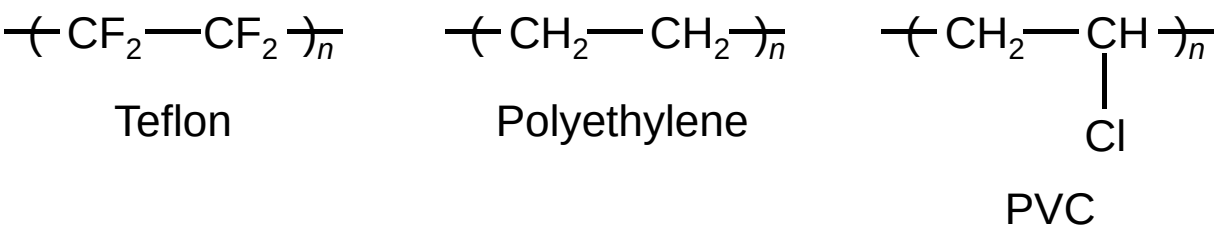
ethylene polyethylene

$$n\mathrm{H_2N(CH_2)_6NH_2} + n\mathrm{HOOC(CH_2)_4COOH} \longrightarrow \mathrm{H\text{-}[NH(CH_2)_6NHCO(CH_2)_4CO]_n\text{-}OH} + n\mathrm{H_2O}$$

나일론-66 축중합반응

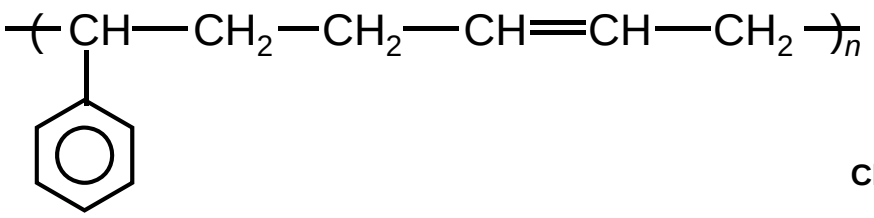
Homopolymers and Copolymers

- ***Homopolymer*** is a polymer made up of only one type of monomer



Teflon Polyethylene PVC

- ***Copolymer*** is a polymer made up of two or more monomers.

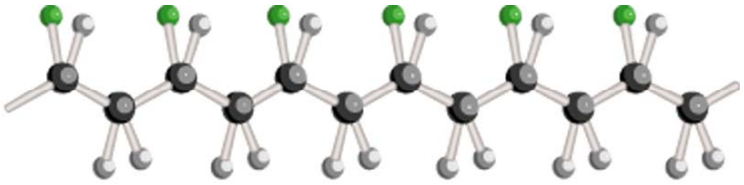


Chang, 25.1

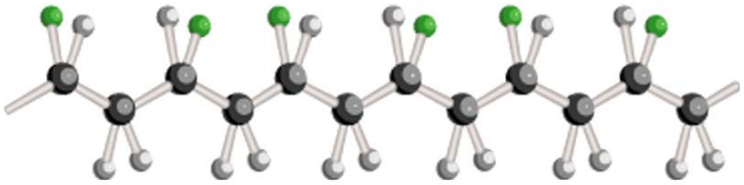
Styrene-butadiene rubber

Stereoisomers of Polymers

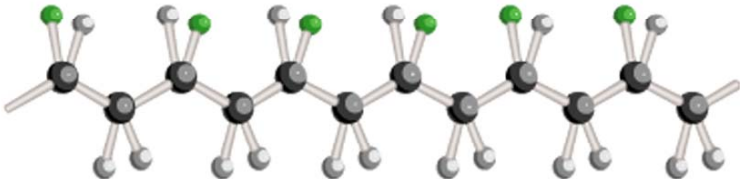
Isotactic: similar side groups always on same side of the chain. Highest melting point and great crystallinity



Syndiotactic: side groups on alternating sides of the chain



Atactic: side groups disposed at random. Rubbery,amorphous, and relatively weak



Common Polymers (1)

TABLE 25.1 Some Monomers and Their Common Synthetic Polymers

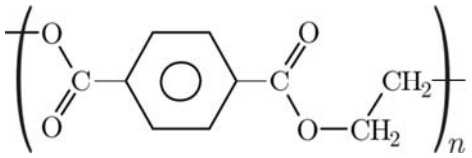
Monomer		Polymer	
Formula	**Name**	**Name and Formula**	**Uses**
$H_2C{=}CH_2$	Ethylene	Polyethylene $-(CH_2{-}CH_2)_n-$	Plastic piping, bottles, electrical insulation, toys
$H_2C{=}C(H)(CH_3)$	Propylene	Polypropylene $-(CH(CH_3){-}CH_2{-}CH(CH_3){-}CH_2)_n-$	Packaging film, carpets, crates for soft-drink bottles, lab wares, toys
$H_2C{=}C(H)(Cl)$	Vinyl chloride	Poly(vinyl chloride) (PVC) $-(CH_2{-}CH(Cl))_n-$	Piping, siding, gutters, floor tile, clothing, toys
$H_2C{=}C(H)(CN)$	Acrylonitrile	Polyacrylonitrile (PAN) $-(CH_2{-}CH(CN))_n-$	Carpets, knitwear

Chang, 25.2

Common Polymers (2)

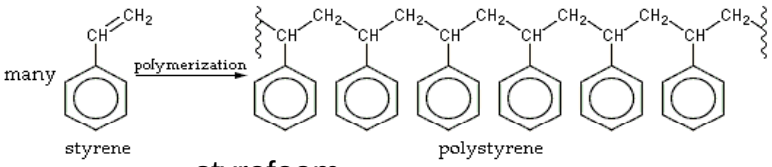
- Polyethylene terephthalate (PET)





- Polystyrene







styrofoam

- High-density polyethylene (HDPE) and low-density polyethylene (LDPE) and



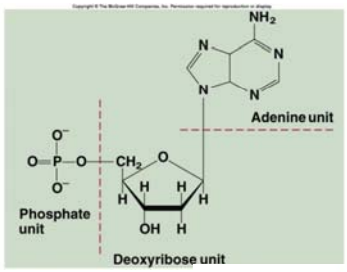




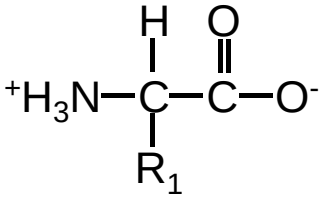


Biopolymers

- **DNA (RNA)** consists of subunits called nucleotides: sugar deoxyribose (ribose), phosphate group PO_4^- and a base.

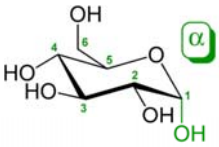


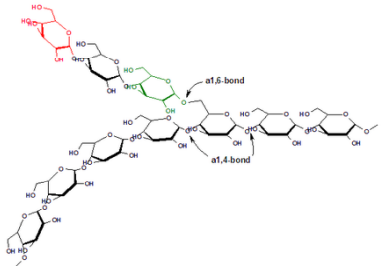
- **Proteins** are polypeptides with 20 different amino acids as the structural subunits.





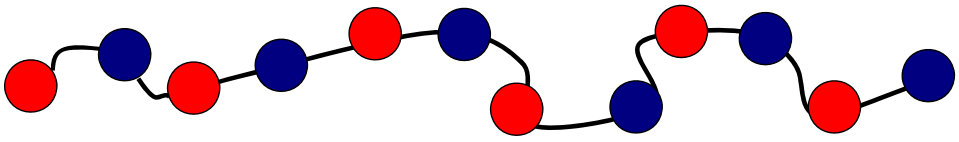
- **Polysaccharides** are complex, often branched, carbohydrates, consisting of monosaccharides (simple sugars)



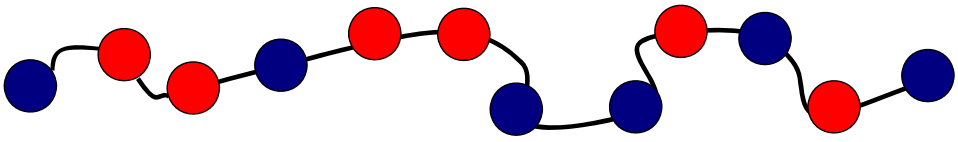


Copolymers (공중합체)

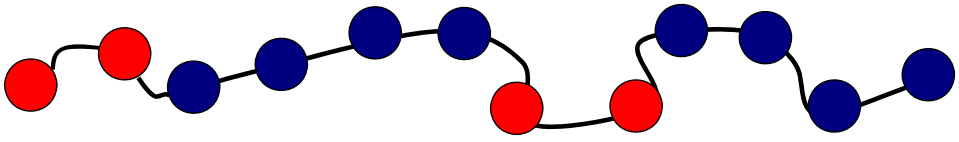
- Alternating copolymers



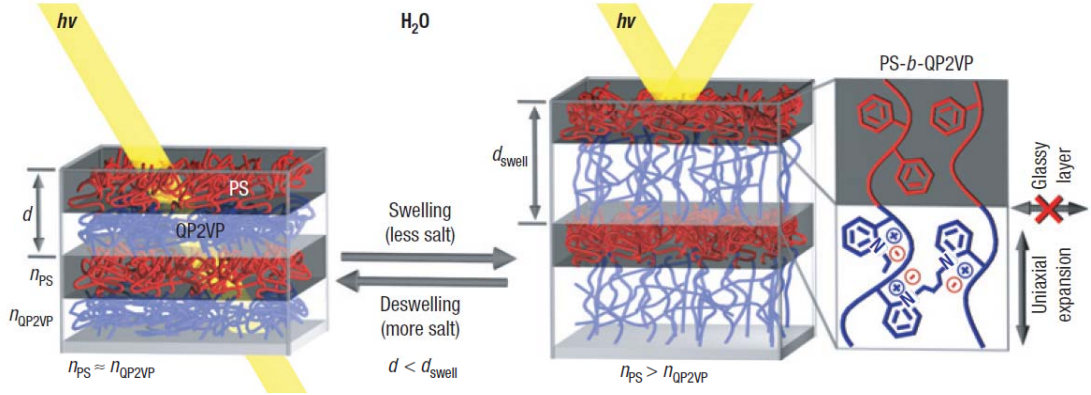
- Random copolymers

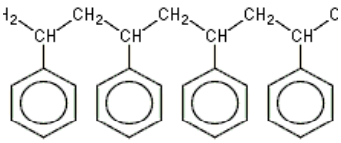


- Block copolymers

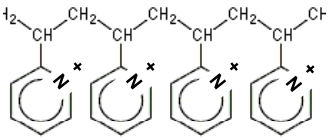


Block-Copolymer Photonic Gel





polystyrene
(=PS)



quarternized poly(2-vinyl pyridine)
(=QP2VP)



Kang et al., Nature Materials **6**, 957 (2007)

고분자의 평균 분자량과 다분산지수

- 수평균 분자량 (number-average molar mass):

$$M_n = \sum_N n_N M_N$$

M_N : N개의 단량체로구성된 고분자의 몰질량
n_N : 몰질량이 M_N인 고분자의 몰분율

- 질량평균 분자량 (weight-average molar mass):

$$M_w = \frac{\sum_N n_N M_N^2}{\sum_N n_N M_N} = \frac{\sum_N n_N M_N^2}{M_n}$$

- 단순분산 (monodisperse): 고분자계를 이루는 모든 고분자들이 같은 수의 단량체로 이루어진 경우
- 다분산 (polydisperse): 고분자계를 이루는 고분자들이 여러 개수의 단량체로 이루어진 경우
- 다분산지수 (polydispersity index)

$$I_p = \frac{M_w}{M_n}$$



(예제) 몰질량이 **1.0×10⁵ g/mol**인 **gelatin monomer** 단백질과 몰질량이 **2.0×10⁵ g/mol**인 **gelatin dimer** 단백질을 포함하는 시료가 있다. **gelatin monomer**의 몰분율이 **0.5**라고 할 때, 시료 속 **gelatin** 단백질의 수평균 몰질량과 질량평균 몰질량을 구하여라. 이 단백질의 다분산지수는 얼마인가?

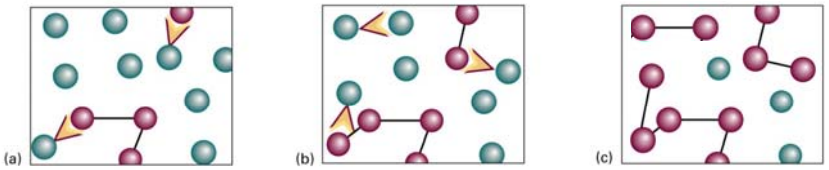
$$\text{수평균 몰질량} = \sum_N n_N M_N = 1.5\times10^5 \text{ g/mol}$$

$$\text{질량평균 몰질량} = \frac{\sum_N n_N M_N^{\,2}}{\sum_N n_N M_N} \cong 1.7\times10^5 \text{ g/mol}$$

$$\text{다분산지수: } I_p = \frac{M_w}{M_n} \cong 1.1$$

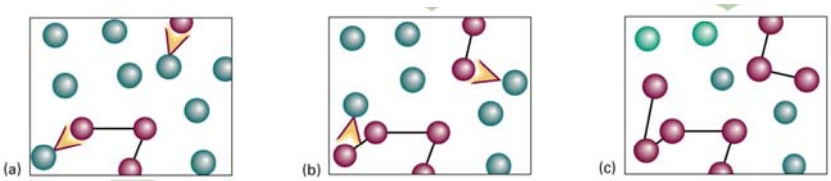
중합반응의 속도론: 단계중합반응과 연쇄중합반응

- 단계중합반응 (stepwise polymerization): 반응 혼합물에 있는 임의의 두 단량체가 서로 결합할 수 있으며, 고분자의 성장은 기존의 형성된 사슬들에만 국한되지 않는다. 고분자 성장속도가 느리다.



$$n\mathrm{H_2N(CH_2)_6NH_2} + n\mathrm{HOOC(CH_2)_4COOH} \longrightarrow$$
$$\mathrm{H\text{-}[NH(CH_2)_6NHCO(CH_2)_4CO]_n\text{-}OH} + n\mathrm{H_2O}$$

- 연쇄중합반응 (chain polymerization): 활성화된 단량체가 다른 단량체와 결합하고, 이 이량체가 다른 단량체와 결합하여 고분자 성장이 일어난다. 고분자 성장속도가 빠르며, 반응시간을 증가시킬수록 긴 사슬의 고분자가 얻어진다.



$$\mathrm{-CH_2CHX\cdot + CH_2{=}CHX \longrightarrow CH_2CHXCH_2CHX\cdot}$$

단계중합반응의 속도법칙

- nylon-66의 단계중합반응

$$n\mathrm{H_2N(CH_2)_6NH_2} + n\mathrm{HOOC(CH_2)_4COOH} \longrightarrow$$
$$\mathrm{H\text{-}[NH(CH_2)_6NHCO(CH_2)_4CO]_n\text{-}OH} + n\mathrm{H_2O}$$

- 단계중합반응의 속도법칙

$$\frac{d[\text{-COOH}]}{dt} = -k[\text{-OH}][\text{-COOH}] = k[\text{-COOH}]^2$$

- 단계중합반응의 적분속도법칙

$$[\text{-COOH}]_t = \frac{[\text{-COOH}]_0}{1 + kt[\text{-COOH}]_0}$$

- *p*: *t* 시간 동안 반응한 작용기 분율

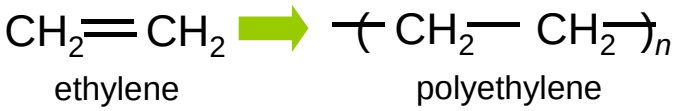
$$p(t) = \frac{\text{시간}t\text{동안 중합반응에참여한작용기수}}{\text{초기작용기수}} = \frac{[\text{-COOH}]_0 - [\text{-COOH}]_t}{[\text{-COOH}]_0} = \frac{kt[\text{-COOH}]_0}{1 + kt[\text{-COOH}]_0}$$

- 중합도 (degree of polymerization, <n>): 한 개의 고분자를 구성하는 평균 단량체 수

$$\langle n(t) \rangle = \frac{[\text{-COOH}]_0}{[\text{-COOH}]_t} = \frac{1}{1 - p(t)} = 1 + kt[\text{-COOH}]_0$$

연쇄중합반응의 반응메커니즘과 속도법칙

- 연쇄중합반응



ethylene　　polyethylene

- 반응메커니즘

1) 개시단계

$$\mathrm{I} \xrightarrow{k_i} \mathrm{R\cdot + R\cdot} \qquad v_i = k_i[\mathrm{I}]$$
$$\mathrm{M + R\cdot} \xrightarrow{k_1} \mathrm{\cdot M_1} \qquad v_1 = k_1[\mathrm{R\cdot}][\mathrm{M}] \quad \text{(fast)}$$

2) 전개단계

$$\mathbf{M + \cdot M_1} \xrightarrow{k_p} \mathbf{\cdot M_2} \qquad \boldsymbol{v_p = k_p[M][\cdot M_1]}$$
$$\mathbf{M + \cdot M_2} \xrightarrow{k_p} \mathbf{\cdot M_3} \qquad \boldsymbol{v_p = k_p[M][\cdot M_2]}$$
$$\vdots$$
$$\mathbf{M + \cdot M_{n-1}} \xrightarrow{k_p} \mathbf{\cdot M_n} \qquad \boldsymbol{v_p = k_p[M][\cdot M_{n-1}]}$$

3) 종결단계

$$\mathrm{M_m\cdot + \cdot M_n} \xrightarrow{k_t} \mathrm{M_{m+n}} \qquad v_t = k_t[\mathrm{M_m\cdot}][\mathrm{\cdot M_n}] \approx k_t[\mathrm{M\cdot}]^2$$

- 연쇄중합반응의 속도법칙 (정류상태근사법 이용)

$$\left(\frac{d[\mathrm{M}\cdot]}{dt}\right) = \left(\frac{d[\mathrm{M}\cdot]}{dt}\right)_{\text{production}} - \left(\frac{d[\mathrm{M}\cdot]}{dt}\right)_{\text{termination}}$$

$$= 2\phi k_i[\mathrm{I}] - 2k_t[\cdot\mathrm{M}]^2 \approx 0 \Rightarrow \boxed{[\mathrm{M}\cdot] = \left(\frac{\phi k_i}{k_t}\right)^{1/2}[\mathrm{I}]^{1/2}}$$

- 단량체 소모의 속도법칙

$$\boxed{\frac{d[\mathrm{M}]}{dt} = -k_p[\mathrm{M}\cdot][\mathrm{M}] = -k_p\left(\frac{\phi k_i}{k_t}\right)^{1/2}[\mathrm{I}]^{1/2}[\mathrm{M}]}$$

- 동적사슬길이 (kinetic chain length):

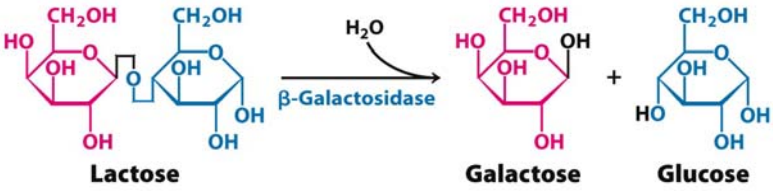
$$\nu = \frac{\text{소모된단량체의개수}}{\text{개시단계에서생성된라디칼개수}} = \frac{|\text{단량체의소모속도}|}{\text{개시단계의라디칼생성속도}}$$

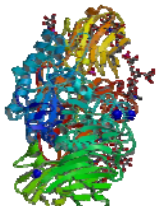
$$\boxed{\nu = \frac{k_p[\mathrm{M}\cdot][\mathrm{M}]}{2\phi k_i[\mathrm{I}]} = \frac{k_p[\mathrm{M}\cdot][\mathrm{M}]}{2k_t[\mathrm{M}\cdot]^2} = \frac{k_p[\mathrm{M}]}{2k_t[\mathrm{M}\cdot]} = \frac{k_p[\mathrm{M}]}{2(\phi k_i k_t)^{1/2}[\mathrm{I}]^{1/2}}}$$

- 연쇄중합법칙의 중합도

$$\langle n\rangle = 2\nu = \frac{k_p[\mathrm{M}]}{(\phi k_i k_t)^{1/2}[\mathrm{I}]^{1/2}}$$

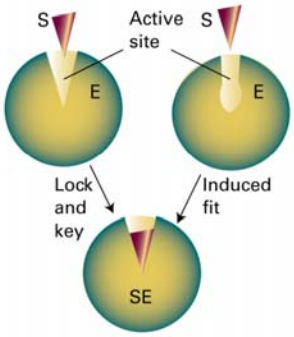
3.2 효소의 촉매반응과 Michaelis-Menten 메컨니즘





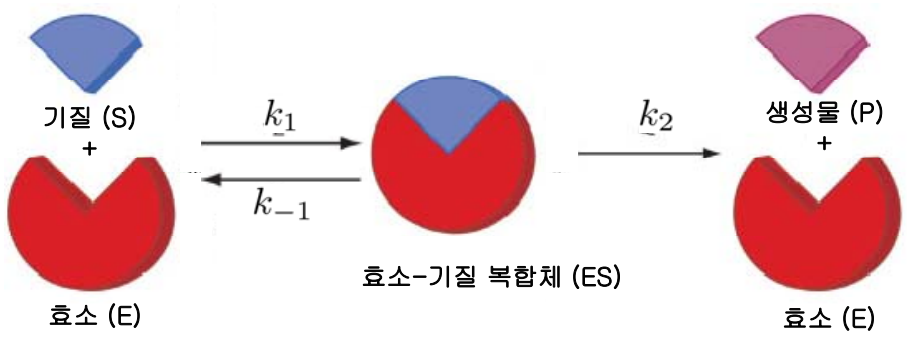
β-galatosidase

- 효소 (enzyme): 생물학적 촉매로서, 일반적으로 단백질로 구성. 활성화 자리에 유기 또는 무기 공동인자가 있음.
- 기질 (substrate): 효소와 반응하는 반응물질
- 효소-기질 반응의 두 가지 모델
  1) 자물쇠-열쇠모형 (key-and-lock model)
  2) 유발적합모형 (induced fit model)



3.2.1 Michaelis-Menten Mechanism

- Michaelis-Menten 메커니즘



$$E+S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES$$

$$ES \xrightarrow{k_2} E+P$$

k_2를 **전환수 (turnover number)**라고 하며, 단위 시간 당 생성물로 전환될 수 있는 기질의 최대 수로서, 일반적으로 1과 10^5 s^{-1} 사이의 값을 갖는다.



- 효소-기질 복합체에 대한 속도법칙과 정류상태근사법

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] \Rightarrow [ES] = \frac{k_1[E][S]}{k_{-1}+k_2} = \frac{[E][S]}{k_M}$$

Michaelis 상수

$[E]_0=[E]+[ES]$이고, 기질의 농도가 매우 높다면 $[S]\approx[S]_0$이므로

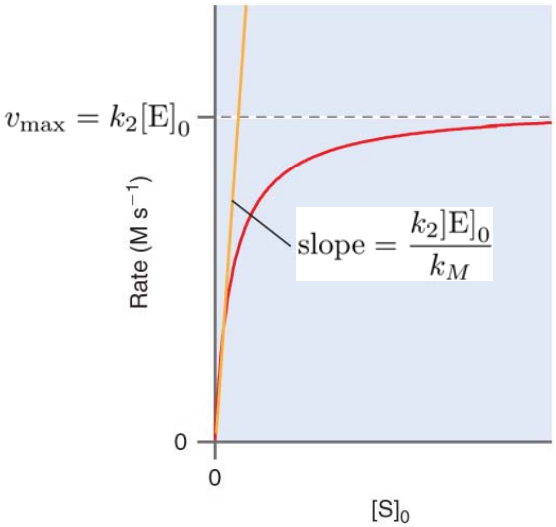
$$[ES] \approx \frac{([E]_0 - [ES])\,[S]_0}{k_M} \Rightarrow [ES] = \frac{]E]_0}{1 + k_M/[S]_0}$$

- 생성물에 대한 속도법칙

$$\frac{d[P]}{dt} = v = k_2[ES] = \frac{k_2]E]_0}{1 + k_M/[S]_0}$$

1) $[S]_0 \ll k_M \Rightarrow v = \frac{k_2]E]_0}{k_M}[S]_0$

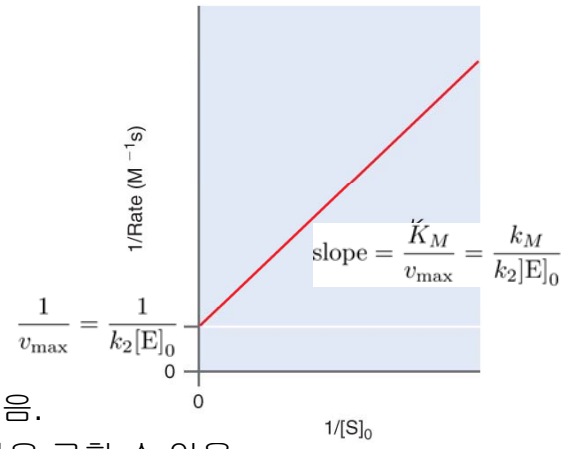
2) $[S]_0 \gg k_M \Rightarrow v = v_{max} = k_2[E]_0$



- Lineweaver-Burk 도표

$$v = \frac{k_2]\mathrm{E}]_0}{1 + k_M/[\mathrm{S}]_0} = \frac{v_{\max}}{1 + k_M/[\mathrm{S}]_0}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{v_{\max}} + \left(\frac{K_M}{v_{\max}}\right)\frac{1}{[\mathrm{S}]_0}$$



– y 절편을 이용하여 k_2의 값을 구할 수 있음.
– k_M의 값은 구할 수 있으나, k_1과 k_{-1}의 값은 구할 수 없음.
– k_1과 k_{-1}의 값은 별도의 실험을 통해 구함.

- 촉매 효율 (catalytic efficiency, ε): 효소의 생성물 생성 효율을 의미

$$\epsilon = \frac{k_2}{k_M} = \frac{k_1 k_2}{k_{-1} + k_2} \leq k_1$$

– $\epsilon_{\max} = k_1$: k_1은 효소와 기질이 결합하는 속도를 의미하고 생체 내에서 이는 확산 속도에 의해 결정.
– 일반적으로 $\varepsilon_{max}=4.0\times10^8$ dm^3 mol^{-1} s^{-1}



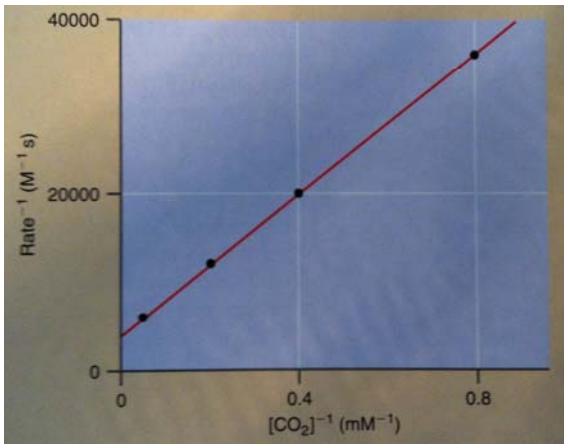
(예제) 근육 등 세포 조직에서 생성된 CO_2는 수화반응에 의해 HCO_3^-로 변환되어 혈관을 통해 폐에 운반된 후, 폐에서 다시 CO_2로 변환되어 배출된다. 이 때 CO_2와 HCO_3^-의 상호 변환에 관여하는 효소가 탄산탈수효소 (carbonic anhydrase)**이며, 이 때 반응식은 다음과 같다.**

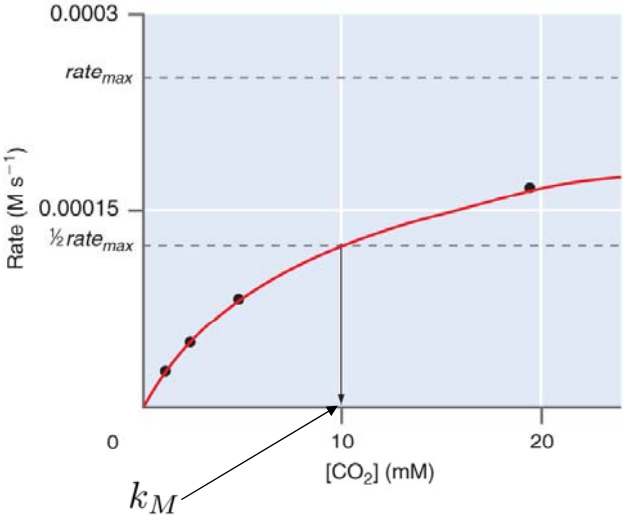
$$CO_2 + H_2O \rightleftarrows HCO_3^- + H^+$$

DeVoe**와** Listiakowsky**는** CO_2 **의 수화반응에 대해 다음과 같은** HCO_3^- **의 생성속도를 얻었다.** (*J. Am. Chem. Soc.* **83**, 274 (1961))

$[CO_2]$ mM	Rate (M s^{-1})
1.25	2.78×10^{-5}
2.50	5.00×10^{-5}
5.00	8.33×10^{-5}
20.0	$1.6\ 7\times 10^{-5}$

초기 탄산탈수효소 농도가 2.3 nM**이라고 할 때,** k_2**와** K_m**의 값을 구하여라.**





$$\Rightarrow \frac{1}{v} = \frac{1}{v_{\max}} + \left(\frac{K_m}{v_{\max}}\right)\frac{1}{[\mathrm{S}]_0}$$

$$v = \frac{k_2]\mathrm{E}]_0}{1 + k_M/[\mathrm{S}]_0} = \frac{v_{\max}}{1 + k_M/[\mathrm{S}]_0}$$